

STAT 515: Final Project - Progress Report

Due before November 22nd, 2024, in-person during office hours. **Hard Copy** must be submitted.

Section Number:	001
Group Number:	07
Names of the Members: (As stated on Blackboard)	Sumanth Veluvolu Sushanth Buddhala Vivek Kumar Koundinya Atchutanna

Title of the Project:	Analyzing and predicting Obesity Through Visualizations and Models
Describe the data set(s) that were considered: (Provide Background information, including the source and the variables)	The dataset we considered is related to obesity statistics among individuals of various age group and genders. The dataset likely originates from a health-related repository, public health organization, further focused on obesity. It captures data including age, gender, height, weight, eating habits (lifestyle), frequency of physical activity, alcohol consumption, usage time of technology devices, modes of transportation, family history of being overweight and obesity levels.
Explain the reasons why the above data set(s) were selected:	We selected this dataset because it addresses a significant and growing concern in the modern world. This dataset deals with obesity and its association with various lifestyle factors.
Research Questions:	1. How can a person with smoking and alcohol consumption tend to be obese be distinguished between genders? 2. What are the trends in BMI obesity rates between males and females, and what gender-specific factors might contribute to these differences? 3. What is the relationship between obesity levels and the amount of time spent on electronic devices? 4. How do physical activity frequency and vegetable consumption frequency influence obesity levels?

Explain the rational behind each question and how questions relate to the data set:	1. • The research question investigates the effects of smoking, alcohol consumption, based on gender on obesity levels. • Dataset has columns related to gender, smoking, alcohol consumption, height, weight and obesity level. Which are used to predict the obesity level. 2. • The goal of the research question is to examine the differences based on gender related to obesity and identifying the underlying factors that might explain these trends. • Dataset has columns related to gender, age, height, weight, with the help of height and weight we can get the BMI index as a new component and use it in predicting the obese level. 3. • This question aims examine whether time spent on electronic devices correlates with obesity levels • For this dataset has columns which say time spent on electronic devices, individuals age, gender, and their obesity levels. 4. • Research question helps to find the how the physical activity frequency and vegetable consumption frequency influence in obesity levels • For this to achieve dataset, the dataset gives the statistics of frequency of physical activity, number of vegetable consumption, obesity levels
Provide an overview of the statistical methods that will be used to answer the research questions:	1. Random Forest: - Analyze the influence of SMOKE , CALC , and Gender on obesity levels (Q1). - Assess how FAF and FCVC predict obesity (Q4). 2. Ridge and Lasso Regression: - Model the impact of SMOKE , CALC , and TUE on obesity (Q1, Q3). - Identify critical predictors like FAF and FCVC (Q4).

	3. Logistic Regression: - Explore gender-specific factors affecting obesity (Q2) - Quantify the impact of TUE , FAF , and FCVC on obesity (Q3, Q4).
Explain the rationale behind the methods used:	1. Random Forest: • Q1 (Smoking, Alcohol, and Gender): Captures complex interactions between Gender , SMOKE , CALC , and obesity levels. It identifies feature importance, which helps distinguish influential variables by gender. • Q4 (Physical Activity and Diet): Handles multiple variables (FAF , FCVC) with non-linear relationships, showing how combinations of these factors influence obesity. 2. Ridge and Lasso Regression: • Q1 (Smoking, Alcohol, and Gender): Lasso selects critical predictors (e.g., SMOKE , CALC) while removing less important ones. Ridge ensures all correlated predictors are accounted for without overfitting. • Q3 (Time on Devices): Helps quantify the impact of TUE while avoiding overfitting if TUE is correlated with other predictors. • Q4 (Physical Activity and Diet): Lasso pinpoints whether FAF or FCVC is more impactful on obesity levels. 3. Logistic Regression • Q2 (Gender-Specific BMI Trends): Provides interpretable coefficients (odds ratios) to understand how predictors like BMI , FAF , and Gender contribute to obesity levels. • Q3 (Time on Devices): Models how TUE influences the probability of obesity while controlling for other predictors. • Q4 (Physical Activity and Diet): Tests the statistical significance of FAF and FCVC on obesity, with coefficients showing the magnitude and direction of their effects.
Any Challenges or issues faced/facing:	
Comments (If any):	

